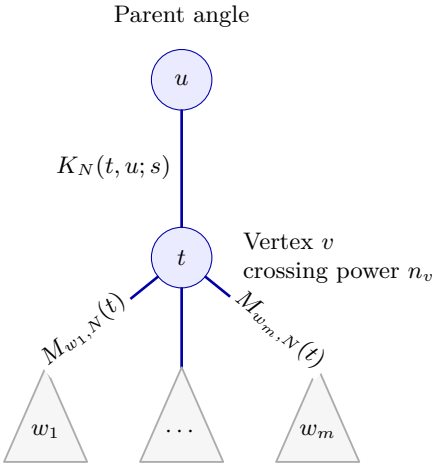


Each child subtree has already been evaluated as a polynomial in $-2\cos t$.

The edge keeps every index $0 \le k < N$:

$$K_N(t,u;s) = \sum_{k=0}^{N-1} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N}{W_k}.$$

The trace at v acts only in t:

$$M_{v,N}(u) = \gamma_{e_v}\,\mathcal{D}_v^{(t)}\left[K_N(t,u;s)\prod_{j=1}^{m} M_{w_j,N}(t)\right].$$

Trigonometric degree in u: at most $N-1$.

Factor out the exact value at zero:

$$\widehat{M}_{v,N}(u) = \frac{M_{v,N}(u)}{M_{v,N}(0)}.$$

Its local phase and its real decay control the next trace. The denominator includes all crossing and orientation factors from this subtree.